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eXplainMR: Generating Real-time Textual and Visual eXplanations to Facilitate UltraSonography Learning in MR

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Introduction (Research Problem)

Teaching cardiac Point-of-Care Ultrasound (PoCUS) is challengeng:

- **Requires visuomotor skills**: moving probe based on visual feedback
- **Requires visuospatial skills**: understanding 3D anatomy from 2D images
- **Requires troubleshooting skills**: recovering from errors

Introduction (Motivation & HCI Importance)

Motivation

- To move beyond simple "follow the arrow" guidance to "meaningful learning" through explanations.
- Learners memorize probe positions without understanding *why.*
- Shortage of qualified ultrasound instructors.
- High cost of simulation equipment.

HCI Importance

- Focuses on Explainable AI (XAI) in Augmented/Mixed Reality.
- Bridging cognitive and physical learning in MR.
- Addressing open-skill tasks.

Background & Related Work

Existing Approaches :

Approach	Strength	Limitation
High-fidelity simulators (Ultrasound Mentor)	3D heart visualization	No intelligent feedback
MR tutoring systems (Adaptutor, CAPtur)	Immersive guidance	Closed-skill, single-path
Subgoal learning (programming, math)	Breaks complex tasks	Not explored in physical-visual tasks

Gap Identified

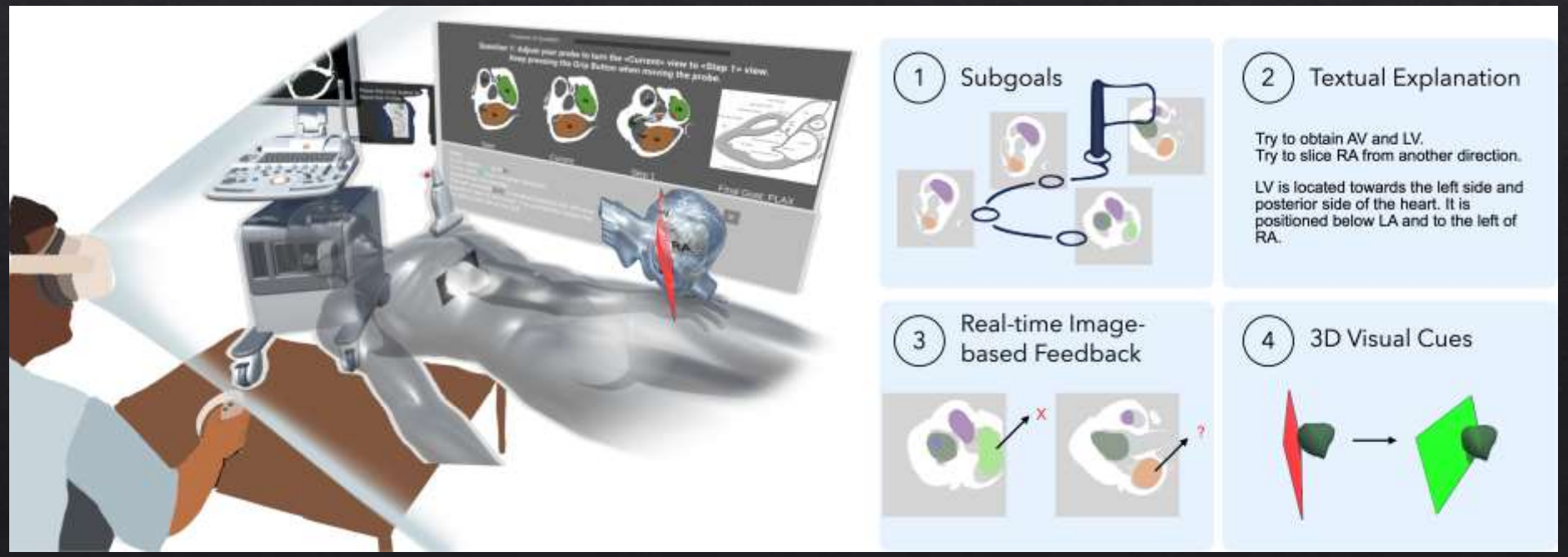
- No MR system provides **real-time explanations** linking probe movement to anatomical changes
- Existing systems tell *how* to move, not *what* is wrong or *why* movement is needed

Proposed Method

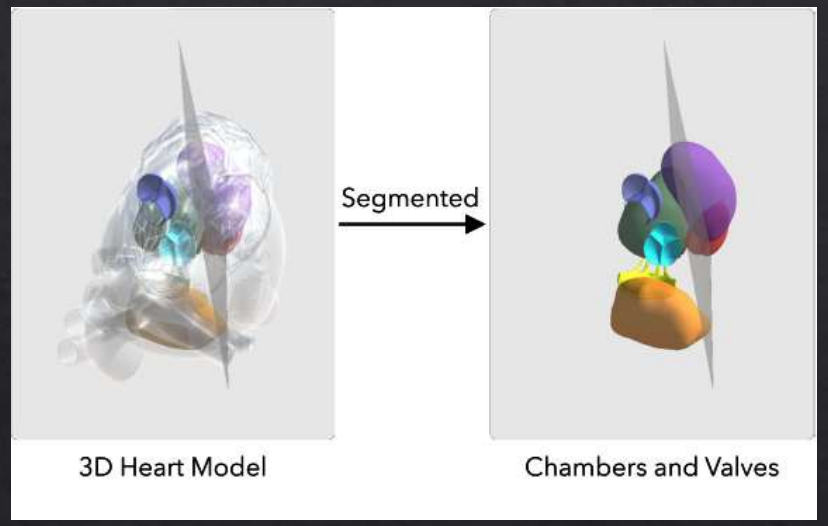
❖eXplainMR

An MR tutoring system that automatically generates:

1. **Subgoals**: Single-movement milestones
2. **Textual Explanations**: What structures are missing or incorrect
3. **Real-time Image Feedback**: Segmented views with labels
4. **3D Visual Cues**: Animated slicing planes and heart models



❖ Architecture

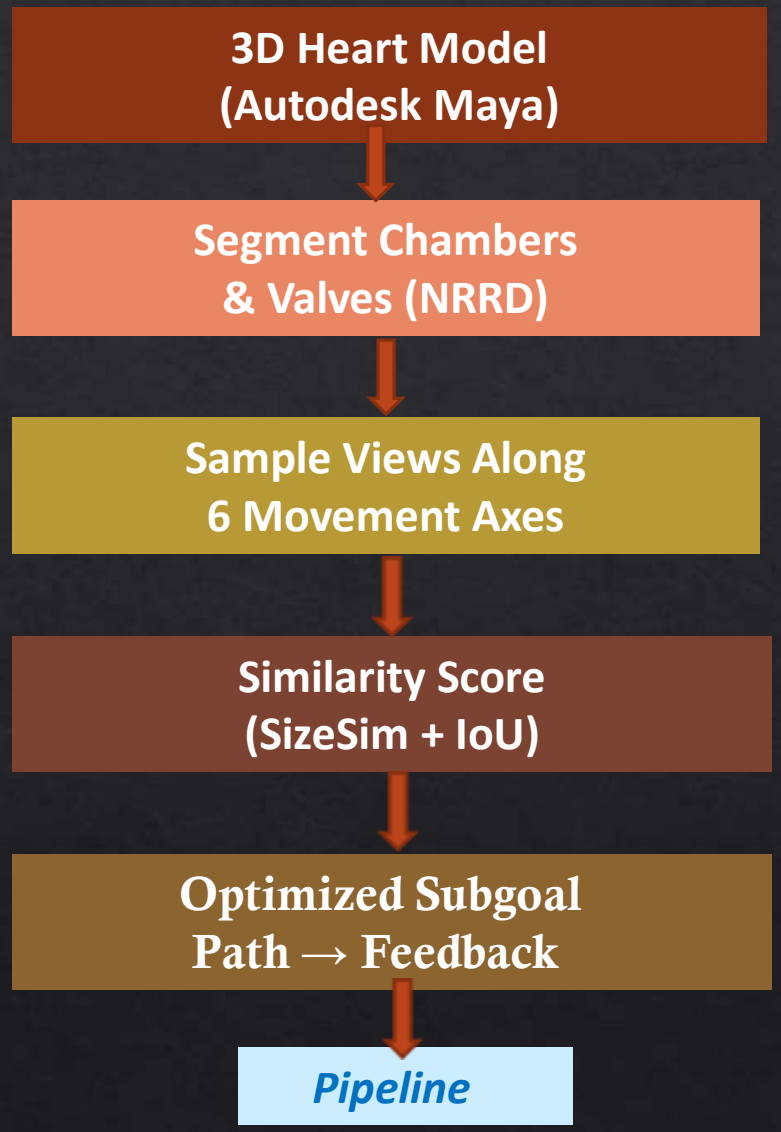


- **3D Heart Model:** Segmented into 4 chambers + 4 valves using Autodesk Maya
- **Subgoal Generation Algorithm:** Optimizes next view based on anatomical similarity
- **Real-time Comparison:** Segmentation maps compared between current and target views

Technologies Used

Component	Technology
Headset	Pico 4 Enterprise
Engine	Unity3D
3D Modeling	Autodesk Maya
Segmentation	3D Slicer

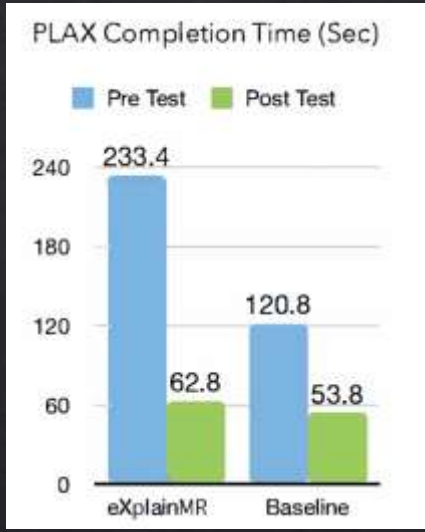
Flowchart

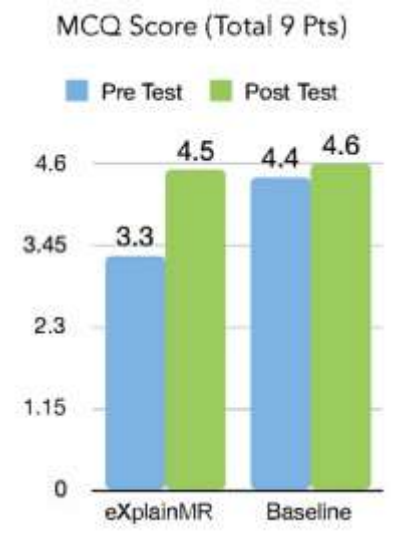

3D Heart Model
(Autodesk Maya)
Segment Chambers
& Valves (NRRD)
Sample Views Along
6 Movement Axes
Similarity Score
(SizeSim + IoU)
Optimized Subgoal
Path → Feedback
Pipeline

Results and Evaluation

Experimental Setup

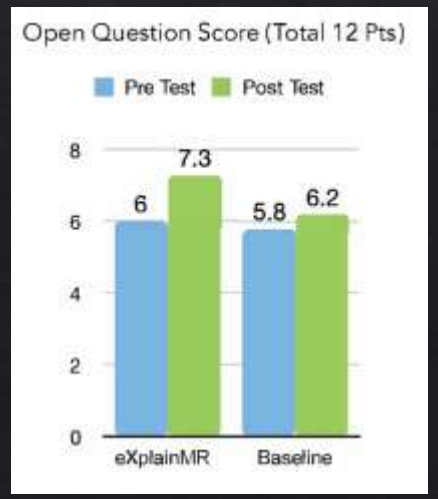
- **Participants:** N=16 medical trainees (students, residents, fellows)
- **Design:** Between-subjects
  - **eXplainMR Group:** Subgoals + Explanations
  - **Baseline Group:** Subgoals + Arrows/Shadows
- **Tasks:** Obtain PLAX view, answer MCQs, explain reasoning
- **Duration:** 1.5 hours (pre-test, training, post-test, interview)





Key Findings

Metric	eXplainMR	Baseline
PLAX time reduction	73.1% ↓	55.4% ↓
MCQ improvement	3.33 → 4.5 (p=0.079)	4.4 → 4.6
Open-ended scores	6.0 → 7.33	6.4 → 7.0



Key Contributions

1 First Work to…

Automatically generate subgoals, textual explanations, and 3D visual cues from a segmented 3D heart model , no human labelling required.

2 Introduces…

An optimization-based subgoal algorithm (SizeSim + IoU) that mirrors expert step-by-step probe navigation rather than naïve 6-DoF decomposition.

3 First MR System to…

Teach open-skill PoCUS by explaining WHY a move is needed (anatomical context) not just HOW to move bridging the expert verbalisation gap.

4 Contributes…

A formative study with 5 expert sonographers identifying the muscle-memory problem, plus a between-subjects N=16 comparative evaluation in a live teaching hospital.

Limitations

Methodological Limitations

- Small sample size (N=16) limits generalizability.
- Short training duration (20 min).
- No real-patient validation .

System Limitations

- MR controller lacks tactile realism of real ultrasound probe.
- Fixed optimal path (may not suit all learners).
- Subgoal granularity (sometimes too many steps, hindering fluidity).

User Experience Limitations

- Cognitive overload for advanced learners (textual explanations redundant).
- Learning curve for multi-step interactions.

Improvement Suggestion

Short-term Improvements

1.Adaptive subgoal generation → reduce/increase steps based on learner performance.

2.Haptic feedback → – simulate tissue resistance using haptic controllers.

3.Personalized explanations → adjust detail level based on expertise.

Long-term Research Directions

4. Learner sourcing → generate multiple troubleshooting strategies from expert demonstrations.

5. AI-powered error prediction → provide proactive hints before user makes mistakes.

6. Longitudinal study → integrate into existing PoCUS curriculum (6 months+).

Conclusion

- eXplainMR successfully bridges the expert-verbalisation gap by auto-generating subgoals, textual comparisons, and 3D animated cues from a segmented heart model.
- Trainees demonstrated faster PLAX completion, more deliberate probe movements, and richer anatomical language — qualitatively favouring explanations over arrows/shadows.
- The optimisation-based subgoal algorithm was validated by 3 expert sonographers as reflecting real-world teaching practice in 80% of cases.
- Quantitative between-group differences were non-significant — larger-scale, longitudinal studies are needed to confirm the learning advantage of explanations.

Future work: Adaptive paths, real-probe haptics, LLM feedback, and generalisation to other organs could make eXplainMR a comprehensive medical MR training platform.

References

- Jingying Wang, Jingjing Zhang, Juana Nicoll Capizzano, Matthew Sigakis, Xu Wang, and Vitaliy Popov. 2025. eXplainMR: Generating Real-time Textual and Visual Explanations to Facilitate UltraSonography Learning in MR. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 18 pages. https://doi.org/10.1145/3706598.3714015